Blood Cell Classification

Final report

Jan Moritz Kupisch

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1 Introduction

Blood cell identification through computer vision represents a critical advancement for hematological diagnostics, where cell density and relative abundance in blood smears serve as key indicators for pathologies such as leukemia. Traditional morphological analysis, while fundamental, remains labor-intensive, prone to inter-observer variability, and heavily dependent on specialized expertise often unavailable in resource-limited settings. Even experienced hematologists demonstrate significant disagreement when classifying morphologically ambiguous cells, underscoring the need for automated systems [Matek et al., 2019].

This project develops an automated blood cell classification system using the Mendeley blood cell dataset, beginning with classical machine learning baseline models (Random Forest, SVM, KNN) to establish performance benchmarks for normal blood cell identification. These foundational models leverage handcrafted features capturing morphological characteristics essential for cell differentiation. Subsequent phases implement a CNN deep learning model in a second step, creating a comprehensive framework that addresses both diagnostic efficiency and research applications in hematological analysis.

2 Image Dataset - Mendeley Normal Blood Cell Dataset

The first dataset comprises 17,092 microscopic images of normal peripheral blood cells obtained from the Hospital Clínic de Barcelona's Core Laboratory [Acevedo et al., 2019]. All images were captured using the CellaVision DM96 digital hematology analyzer at a resolution of 360 × 363 pixels in RGB JPG format. Blood samples were collected from donors without infections, hematologic or oncologic diseases, and pharmaceutical treatments at the time of acquisition. Smears were stained using the standardized May-Grünwald-Giemsa protocol, ensuring consistent morphological visualization across samples. The dataset encompasses eight morphological classes representing normal leukocyte populations and precursors and with a moderately imbalanced distribution. Class sizes range from 7.1 % (lymphocyte) do 19.48 % (neutrophil) of the total dataset. This distribution reflects typical clinical blood sample compositions where neutrophils predominate and rare cells like lymphocytes and basophils are less frequent (Fig. 1 and Tab. 1).

1. Neutrophils (segmented and band forms)
2. Eosinophils
3. Immature granulocytes (metamyelocytes, myelocytes, and promyelocytes)
4. Platelet
5. Erythroblasts
6. Monocytes
7. Basophils
8. Lymphocytes (typical)

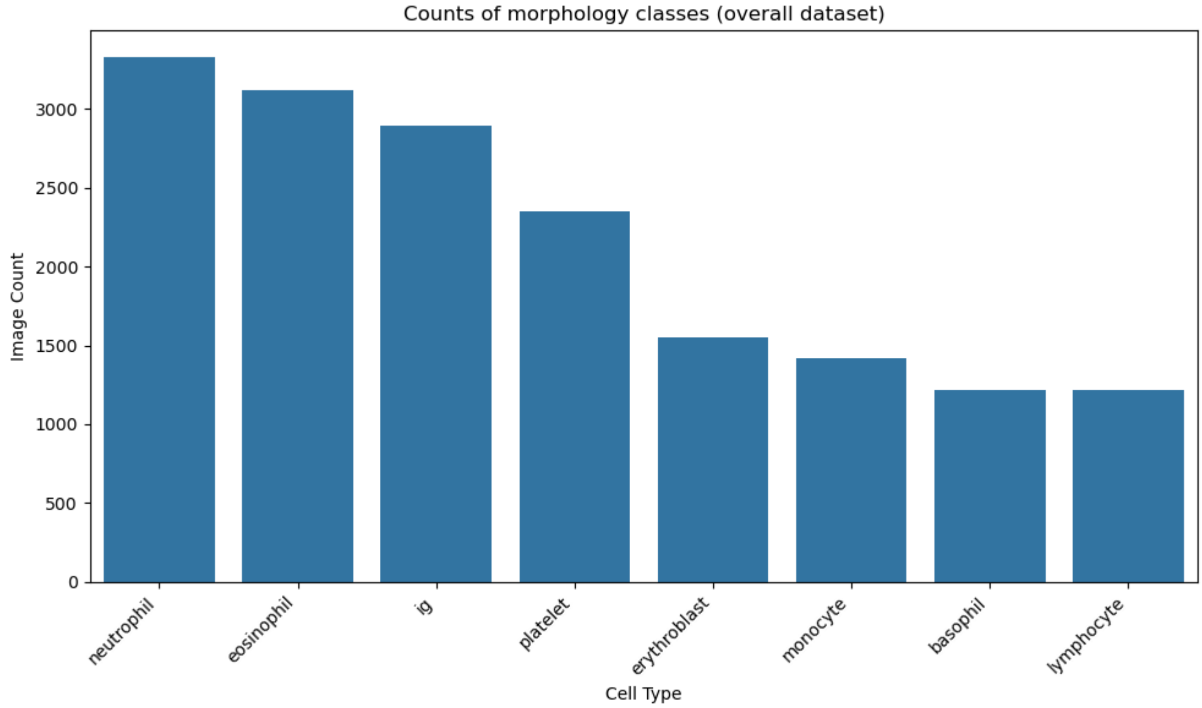

Figure 1: Ordered countplot of the eight different morphology classes in the Normal Blood Cell Dataset.

cell type	count	percentage
neutrophil	3329	19.48
eosinophil	3117	18.24
ig	2895	16.94
platelet	2348	13.74
erythroblast	1551	9.07
monocyte	1420	8.31
basophil	1218	7.13
lymphocyte	1214	7.10

Table 1: Counts and percentages of the eight different morphology classes in the Normal Blood Cell Dataset.

2.1 Image Resolution Distribution

The analysis of image resolutions shows that the dataset contains a variety of image sizes (Fig. 2 and Tab. 2). Where the images have many different width–height combinations and most are around 360×360, but not all. This variety in images sizes means that some pre-processing is necessary. Therefore, it is recommended to resize all images to a standard resolution while keeping their aspect ratios, to maintain consistency and avoid any distortion. Moreover, considering the image resolution statistics:

Mean width: 360.09 pixels
Mean height: 363.05 pixels

Standard deviation of width: 0.72 pixels
Standard deviation of height: 0.79 pixels

We confirm that the average image resolution was found to be 360.09 × 363.05 pixels. The standard deviations of width and height were 0.72 pixels and 0.79 pixels, respectively, indicating extremely low variability in image dimensions. This suggests that the images are nearly uniform in size. Consequently, resizing all images to a fixed resolution is statistically justified and is unlikely to introduce geometric distortion or loss of relevant cellular features.

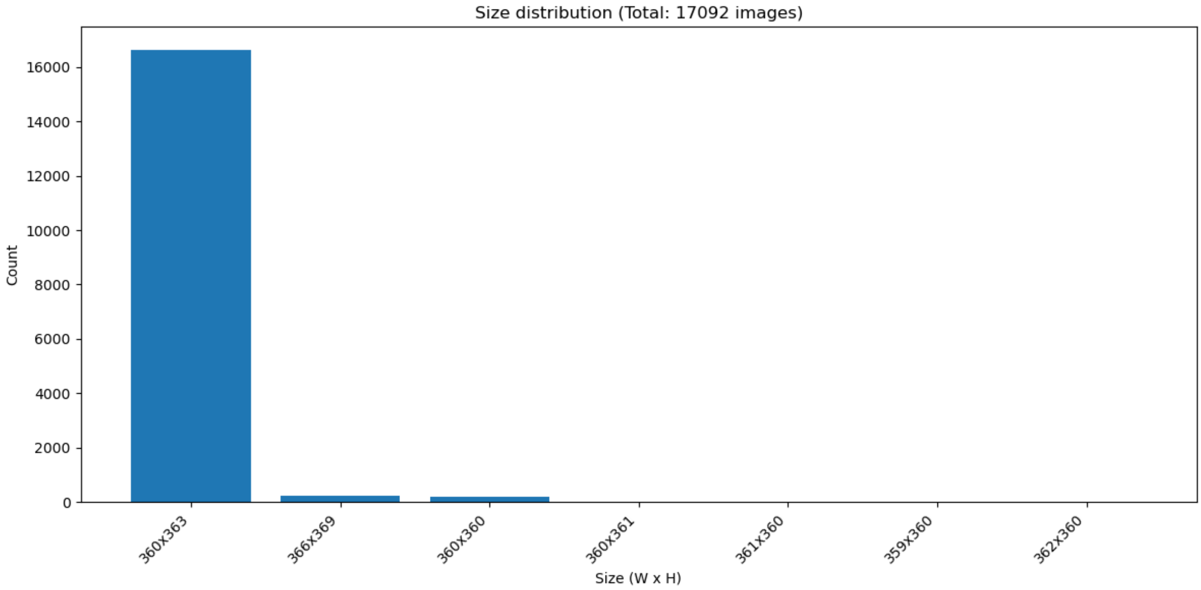

Figure 2: Image size distribution plot.

Width	Height	Count
360.0	363.0	16639
366.0	369.0	250
360.0	360.0	198
360.0	361.0	2
359.0	360.0	1
361.0	360.0	1
362.0	360.0	1

Table 2: Counts of image size distribution.

All images were uniformly resized to 300×300 pixels using centered cropping (Fig. 3) to ensure consistent input dimensions required for machine learning models (SVM, RF, KNN) and subsequent CNN training, despite the dataset containing 97.35% images at 360×363 pixels alongside 7 minor size variants totaling less than 3%.

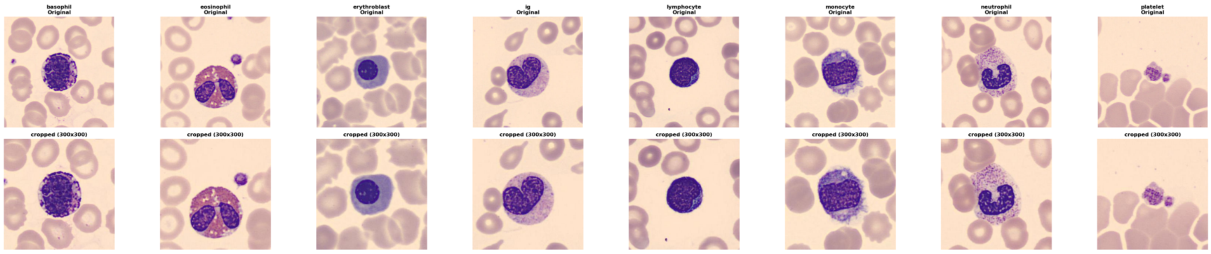

Figure 3: Example plots of eight cell types in the original resolution (upper row) and after cropping to 300 x 300 (lower row).

2.2 Data Quality Assesment

The dataset is arranged so that each folder represents a different types of blood cells and that all images inside a folder share the same label. This structure ensures that the dataset is fully labeled and suitable for supervised learning. A detailed data quality check was carried out, leading to the following observations:

- No missing or incorrect labels were found.
- All image files paths were valid and accessible.
- No corrupted image files were detected.
- Only one image could not be read during resolution analysis and was removed from the dataset.

Overall, the dataset is clean and well-organized.

2.3 RGB and Grayscale Intensity Analysis

To understand the characteristics of different cell types in the datasets, we perform fundamental image analysis based on intensity and color distributions. Grayscale analysis provides insights into overall brightness and contrast patterns by converting RGB images to luminance values using the standard ITU-R BT.601 weighting (0.299R + 0.587G + 0.114B). Channel-specific RGB statistics reveal color variations that may be associated with different staining properties or morphological features of cell types. These basic metrics serve as a foundation for quality control, helping to identify potential imaging inconsistencies, batch effects or significant distributional differences between classes that could inform preprocessing decisions for the CNN classifier.

The mean RGB channel values reveal relatively consistent color profiles, with all classes showing red channel dominance (approximately 215-235) followed by green (185-202) and blue (183-190) channels (Fig. 4). Erythroblast and Platelet exhibit the highest red channel intensities (~232 and ~235 respectively), while ig displays the lowest overall intensity across all channels. The grayscale analysis (Fig. 5) demonstrates relatively uniform brightness levels (193-209) across most cell types. Platelet shows the highest mean brightness (~209) with distinctly lower contrast (~27), while ig, monocyte and basophil show slightly reduced brightness (~195-197) with higher contrast values (~47-48). Overall, the relatively narrow range of color and brightness values suggests that the CNN classifier will likely rely primarily on texture and morphological features rather than color information for differentiation.

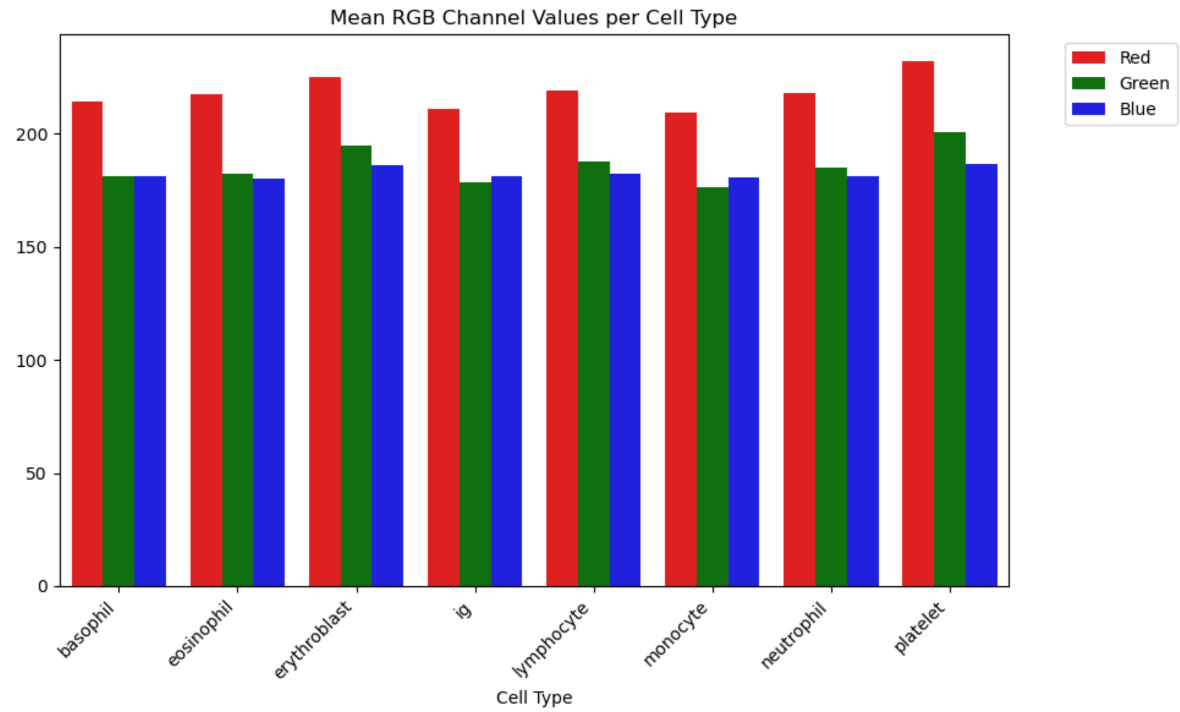

Figure 4: Mean RGB channel values per cell type

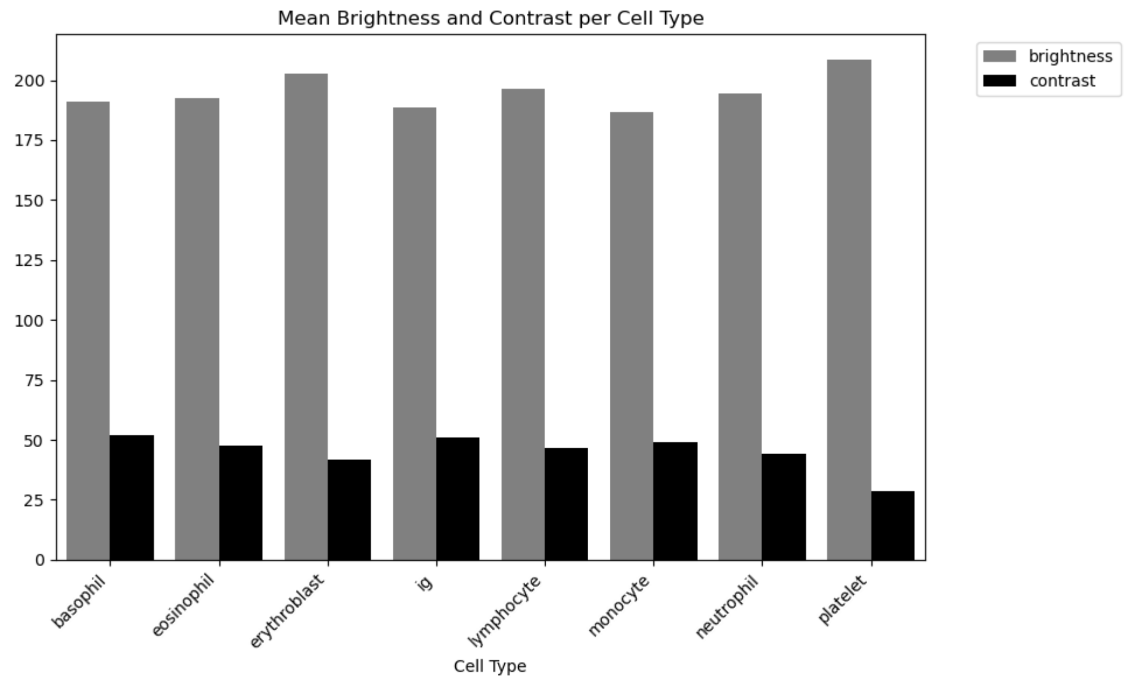

Figure 5: Mean brightness and contrast per cell type.

2.4 Distribution Analysis with Boxplots

The boxplot visualizations show that most cell types display relatively compact interquartile ranges with medians around 195-210, though all classes exhibit outliers (Fig. 6). Ig displays numerous low-intensity outliers extending down to ~155 and exceptionally high contrast variability with outliers extending above 60. Platelet demonstrates the narrowest contrast distribution with a median around 27, confirming its more homogeneous internal structure. For the RGB channels, ig exhibits numerous outliers spanning from ~175 to above 250 in the red channel, while platelet shows elevated green median values (~202). The prevalence of outliers across all metrics suggests some degree of imaging variability that could potentially be addressed through outlier filtering or normalization strategies during preprocessing. The boxplot analysis indicates that the dataset is suitable for CNN training, with most cell types displaying compact interquartile ranges and consistent median values. While outliers are present across all classes, they display consistent directionality across cell types, with 9 outliers occurring predominantly in the same tail of the distribution. A pragmatic approach would be to proceed with training on the complete dataset and only investigate outlier filtering if specific classification problems emerge during model evaluation.

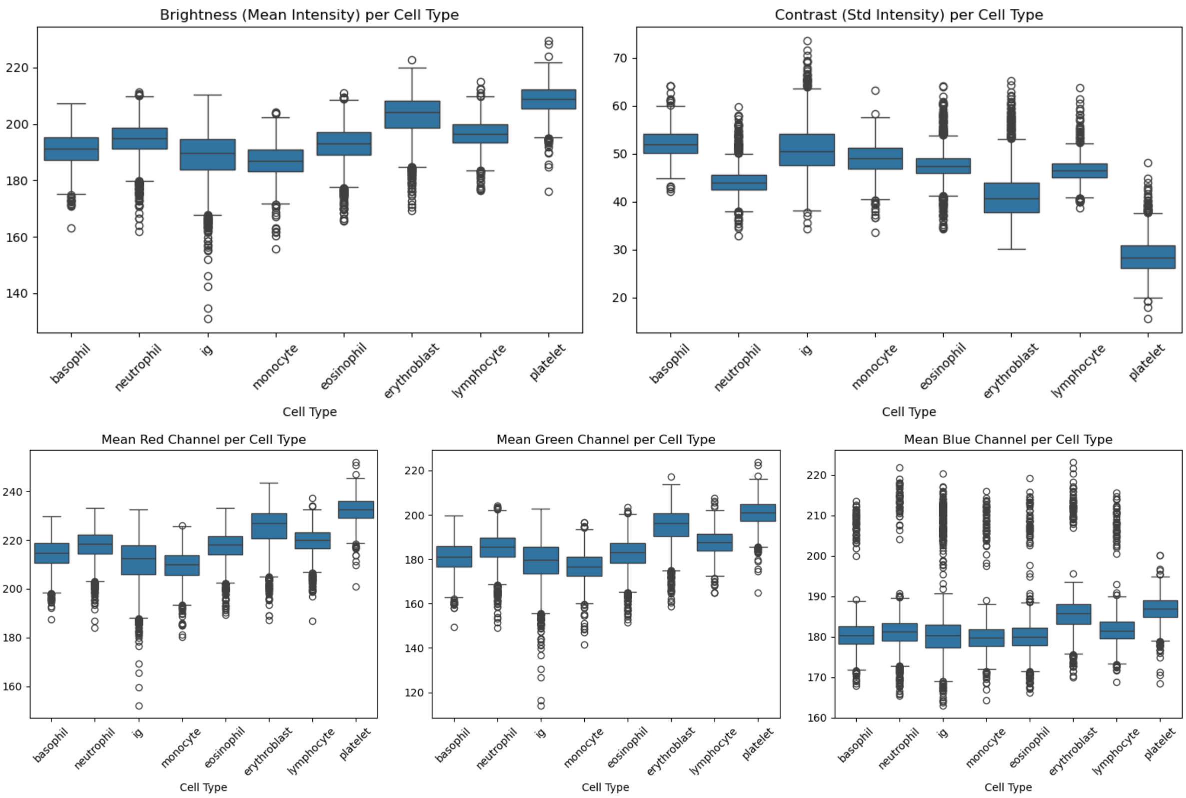

Figure 6: Boxplots of Grayscale and RGB channel distributions per cell type.

3 Feature Engineering

3.1 Train/Validation/Test - Splitting

An 80/10/10 stratified split was employed to ensure robust model training, hyperparameter tuning, and unbiased final evaluation. The training set (80%) trains the models and determines optimal hyperparameters via cross-validation, the validation set (10%) selects the best-performing model configuration among candidates, and the test set (10%) provides an unbiased estimate of a final test of the selected model.

Instead of physically moving images into separate train/val/test subfolders, a DataFrame-based approach preserved the original class folder structure (Data/NormalCells/cropped_pictures/cell_type/). The create_dataset_dataframe() function systematically catalogs all images with their file paths, filenames, and class labels (derived from folder names), automatically encoding labels as integers via pd.Categorical().codes. Subsequent stratified splits using train_test_split(df, stratify=df['label']) create train_df, val_df, and test_df containing file paths and corresponding labels while maintaining class proportions across all splits. This approach enables flexible feature extraction workflows without disrupting the original directory organization.

3.2 Feature extraction

Hand-crafted features capturing color, texture, and shape were extracted to establish interpretable baselines for blood cell classification. Color features (8 total: mean/std of grayscale + RGB channels) quantify staining characteristics critical for cell type identification (nucleus blueness, cytoplasm pinkness). Texture features (3 GLCM properties: contrast, homogeneity, energy) capture cellular granularity and membrane patterns using Gray-Level Co-occurrence Matrices computed on 32 quantized gray levels across horizontal/vertical directions. Shape features (4: normalized area, circularity, aspect ratio, compactness) distinguish morphological differences (erythrocytes: round/small vs. monocytes: larger/irregular) via contour analysis with Gaussian blur preprocessing and Otsu thresholding.

Images were resized to 100×100 pixels using Lanczos interpolation to balance computational efficiency with sufficient resolution for blood cell details, reducing 300×300 originals by 9× while preserving discriminative patterns for classical ML models (SVM/RF/KNN) and CNN preprocessing.

Feature matrices were constructed by iterating over the pre-split DataFrames (train_df, val_df, test_df), applying extract_features() to each image path, and collecting 15-dimensional vectors into NumPy arrays (X_train_simple: 13,673×15, X_val_simple: 1,709×15, X_test_simple: 1,710×15). This preserves the stratified 80/10/10 split while maintaining original folder structure, enabling model training with standardized inputs. Final DataFrames combining features with both numeric (label_id) and categorical (label) identifiers using a label mapping dictionary were created for each split and saved as CSV files (features_train_simple.csv, features_val_simple.csv, features_test_simple.csv) for loading into separate modeling notebooks.

4 Model Development and Results

4.1 Feature Space Analysis (PCA)

4.1.1 3D Feature Space Visualization

Principal Component Analysis (PCA) was performed to validate the discriminative quality of the 15 extracted features and visualize class separability in reduced-dimensional space. The first three principal components capture 82.7% of total variance (PC1: 51.4%, PC2: 20.3%, PC3: 11.0%), indicating that the handcrafted features encode substantial discriminative information despite dimensionality reduction from 15 to 3 dimensions.

The 3D PCA projections (Fig. 7) reveal consistent clustering patterns across all three data splits (train/validation/test), confirming the stratified splitting strategy preserved class distributions effectively. Platelet (cyan) and eosinophil (orange) form the distinct clusters for example, occupying well-separated regions in feature space with minimal overlap, reflecting their characteristic morphological differences (platelets: small/compact vs. eosinophil: larger/lobulated nuclei). Erythroblast (green) exhibits moderate separation along PC1, correlating with its unique staining properties and developmental stage markers.

In contrast, basophil (blue), lymphocyte (brown), monocyte (gray), and immature granulocyte (ig, purple) display substantial overlap in the projected space, particularly along PC2-PC3 axes. This overlap reflects the morphological similarity among mature populations that share granular cytoplasm and comparable nuclear-to-cytoplasmic ratios. The consistent overlap patterns across train/validation/test splits demonstrate that observed class confusion stems from inherent biological similarity.

The preservation of cluster structures across splits validates both the feature extraction pipeline's robustness and the appropriateness of the 80/10/10 stratified division for subsequent model training and evaluation.

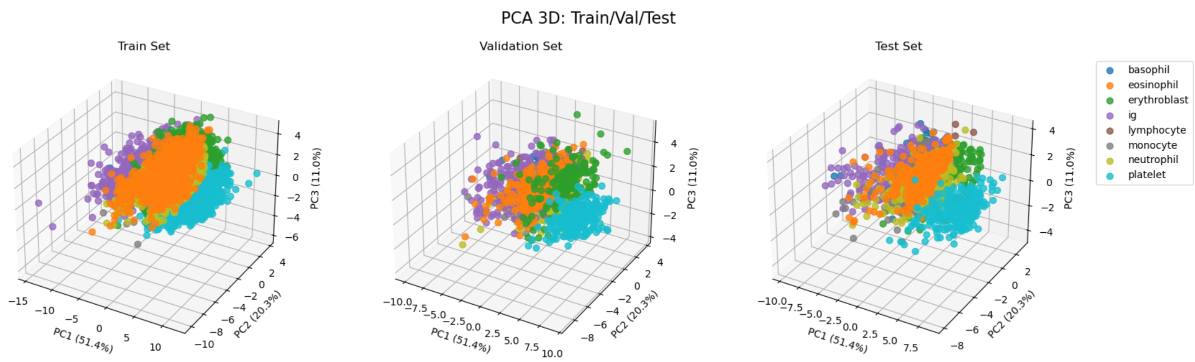

Figure 7: 3D PCA projections of train-, validation-, and test-dataset

4.1.2 PCA Loading Analysis

The PCA loading heatmap (Fig. 8) quantifies individual feature contributions to each principal component, revealing the underlying morphological and staining characteristics that drive class separability. PC1 (51.4% variance) is dominated by color intensity features, with near-equal loadings for grayscale mean (0.33), green channel mean (0.33), and green channel standard deviation (0.33). This pattern indicates that PC1 primarily captures overall cell brightness and green-channel staining intensity variations, which correlate strongly with May-Grünwald-Giemsa uptake differences between cytoplasm-rich cells (monocytes/neutrophils: darker) versus cytoplasm-poor cells (lymphocytes: lighter). The dominance of green-channel features aligns with the eosinophilic staining component in MGG protocols, where cytoplasmic basophilia manifests predominantly in green-shifted hues.

PC2 (20.3% variance) exhibits strong loadings exclusively from shape descriptors: normalized area (0.53), compactness (0.53), and circularity (0.48). This component differentiates cells by geometric properties, separating small/round platelets from larger/irregular monocytes and band neutrophils. The near-zero loadings of color and texture features on PC2 confirm orthogonality between chromatic properties (PC1) and morphological shape (PC2), suggesting these feature categories capture complementary aspects of cellular identity.

PC3 (11.0% variance) is primarily driven by texture energy (0.53), blue channel standard deviation (0.42), and red channel standard deviation (0.28). Texture energy quantifies local pixel homogeneity via GLCM analysis, distinguishing cells with uniform cytoplasm (erythrocytes, lymphocytes) from those with granular inclusions (basophils, eosinophils).

The loading distribution demonstrates that while color features dominate overall variance (PC1: 51%), shape and texture features contribute essential orthogonal information (PC2+PC3: 31%), justifying the multi-modal feature engineering strategy. The relatively low contribution of aspect ratio (0.01–0.10 across all PCs) indicates that cell elongation provides minimal discriminative value for this dataset.

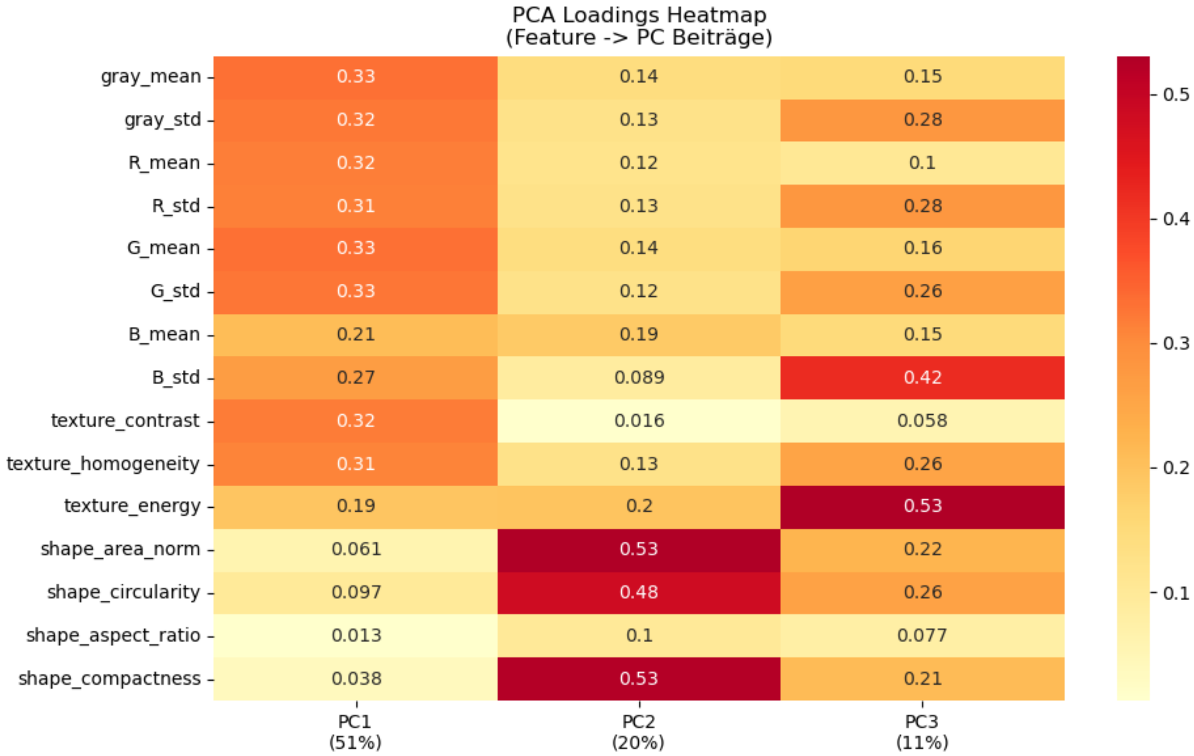

Figure 8: Heatmap of PCA loading quantifying individual feature contributions to each principal component.

4.2 Baseline Model Comparison

4.2.1 Hyperparameter Tuning (Grid Search on Validation Set)

Three classical machine learning algorithms – Random Forest (RF), Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) – were evaluated as baseline classifiers. Hyperparameter tuning employed 5-fold cross-validation grid search on the training set, with final model selection based on validation set performance to prevent overfitting to training data.

Model	**Parameters**	**Typical values**
Random Forest	n_estimators, max_depth	100-500, 10-30
SVM	C, kernel, gamma	0.1-100, rbf/linear, scale/auto
KNN	n_neighbors, weights	3-15, uniform/distance

Table 3: Parameter set and ranges for hyperparameter tuning of the baseline models (RF, SVM, KNN).

The search grids encompassed key architectural parameters: RF explored ensemble sizes (100–300 trees), tree depths (10, 20, unrestricted), and split criteria (2–5 samples); SVM tested regularization strengths (C: 0.1–10), RBF kernel bandwidths (γ: scale/auto/0.001/0.01), and kernel types (RBF/linear); KNN varied neighborhood sizes (k=3–10), distance weighting schemes (uniform/distance), and metrics (Euclidean/Manhattan).

Optimal configurations achieved cross-validation accuracies of 80.7% (RF: 300 trees, unrestricted depth, min_samples_split=2), 84.2% (SVM: C=10, γ=scale, RBF kernel), and 77.4% (KNN: k=10, distance-weighted, Euclidean). Validation set evaluation confirmed SVM superiority (84.0%), outperforming RF (80.2%) and KNN (78.6%) by 3.8–5.4 percentage points, establishing SVM as the lead model for comprehensive performance analysis.

4.2.2 Overall Test Accuracy Comparison (RF vs. SVM vs. KNN)

Final test set evaluation (Fig. 9, left panel) demonstrates that SVM achieves the highest accuracy (84.4%), followed by Random Forest (81.3%) and KNN (78.1%). Random Forest's intermediate performance (81.3%) suggests that ensemble averaging of axis-aligned decision trees partially mitigates overfitting (training accuracy: 100.0%) but fails to match SVM's generalization capability (training: 87.1%, test: 84.4%). The 2.9% train-test gap for SVM versus 18.7% for Random Forest indicates superior regularization through soft-margin optimization, critical for this morphologically ambiguous classification task.

All subsequent analyses focus on the SVM model as the best-performing baseline classifier.

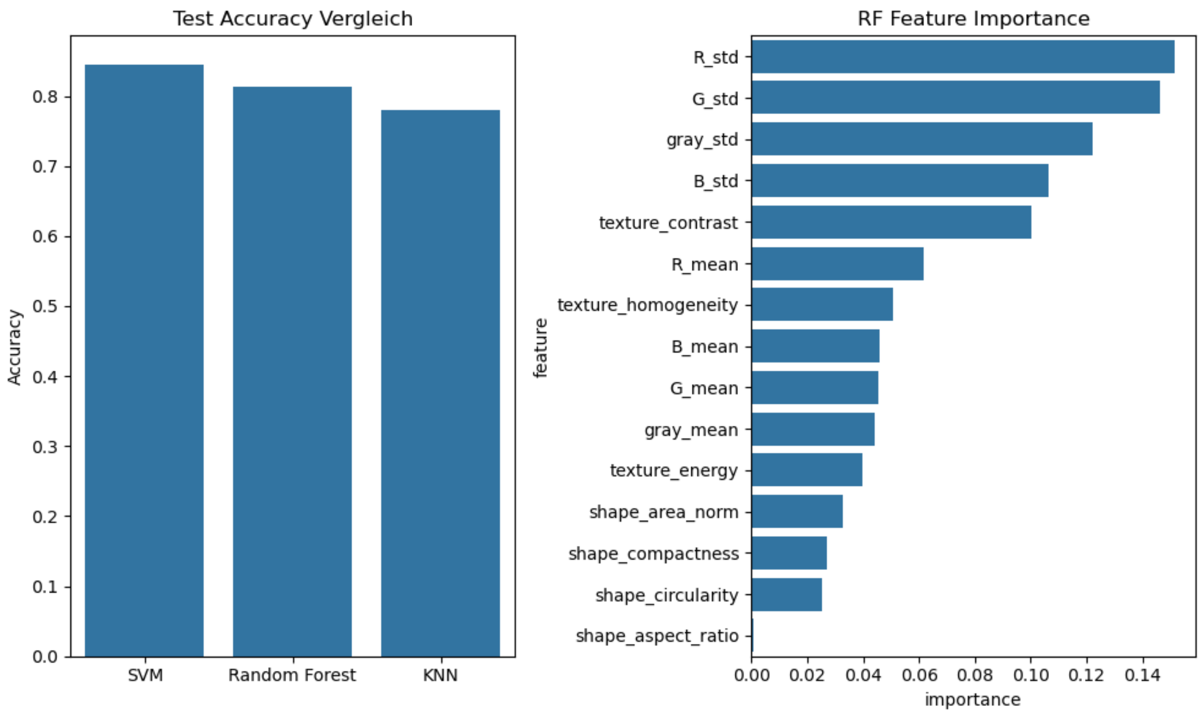

Figure 9: Overall test accuracy comparison of the three baseline models SVM, RF and KNN (left). Random Forest feature importance rankings (right).

4.3 Random Forest Analysis

4.3.1 Feature Importance

Random Forest feature importance rankings (Fig. 9, right panel) reveal that chromatic variability features dominate decision-making: RGB standard deviations (R_std: 0.148, G_std: 0.144, B_std: 0.113) collectively contribute 40.5% of total importance, significantly outweighing their mean counterparts (R/G/B_mean: 5.6–7.0%). This pattern indicates that Random Forest exploits staining heterogeneity – granule distributions, cytoplasmic vacuolization, nuclear chromatin clumping – rather than absolute color intensity for cell type discrimination.

Texture features rank moderately: contrast (0.112) and homogeneity (0.056) capture granular patterns distinguishing basophils/eosinophils from agranulocytes, though their combined 16.8% contribution suggests limited incremental value beyond color variance. Shape descriptors exhibit minimal importance (area_norm: 0.032, circularity/compactness/aspect_ratio: <0.02), contradicting PC2's high variance (20.3%) driven by these features.

The dominance of chromatic over morphological features in Random Forest contrasts with SVM's balanced utilization (evidenced by comparable performance despite shape features' low RF importance), highlighting algorithmic biases in feature utilization and justifying multi-model comparison rather than reliance on a single classifier architecture.

4.4 SVM Detailed Performance Analysis

4.4.1 Classification Report (Precision/Recall/F1-Score)

The SVM classifier achieves 84.4% overall test accuracy with balanced performance across precision (84%) and recall (84%), indicating minimal bias toward majority classes despite the moderately imbalanced dataset (neutrophil: 19.5% vs. lymphocyte: 7.1% of samples). Per-class F1-scores range from 0.68 (ig) to 0.99 (platelet), revealing substantial performance heterogeneity across cell types (Fig. 10).

Platelet classification demonstrates near-perfect performance (precision: 1.00, recall: 0.99, F1: 0.99), attributed to its distinctive morphological signature – small size, high circularity, uniform cytoplasm – that manifests as well-separated clusters in PCA space (Fig. 7, cyan points). Eosinophil (F1: 0.89) and neutrophil (F1: 0.87) achieve strong performance through characteristic granulation patterns captured by texture features (contrast, homogeneity) and chromatic properties (R/G_std importance: 14.8%/14.4%).

In contrast, immature granulocyte (ig) exhibits the lowest F1-score (0.68, precision: 0.69, recall: 0.67), reflecting its transitional developmental stage. This morphological ambiguity – variable nuclear-to-cytoplasmic ratios, inconsistent granule maturation – produces feature overlap with neutrophils (31 misclassifications), eosinophils (21), and monocytes (18), consistent with PCA's purple cluster dispersion (Fig. 7). Similarly, lymphocyte (F1: 0.80) and monocyte (F1: 0.78) suffer from agranulocyte similarity: both lack cytoplasmic granules, exhibit round-to-oval nuclei, and display comparable staining intensities, challenging feature-based discrimination.

The macro-averaged F1-score (0.84) matches weighted-average and overall accuracy, demonstrating that class imbalance does not systematically bias performance, as platelet's superior metrics stem from inherent discriminability rather than sample size (n=235, 13.7% of test set).

Figure 10: Per class performance metrics (Precision, Recall, F1-Score).

4.4.2 Confusion Matrix Analysis

The absolute confusion matrix (Fig. 11, left panel) reveals systematic misclassification patterns driven by morphological similarity. Neutrophil demonstrates robust classification (299/333 correct, 89.8% recall) with principal errors distributed across ig (18), eosinophil (5), and erythroblast (7), reflecting shared granulocyte lineage features (lobulated nuclei, granular cytoplasm, comparable cell size). Conversely, ig exhibits the highest absolute error count (95/290 misclassified, 32.8% error rate), predominantly confused with neutrophil (31).

The normalized confusion matrix (Fig. 11, right panel) quantifies class-specific error distributions. Basophil misclassifications disproportionately favor ig (12.3%) and eosinophil (1.6%), likely due to shared dense granulation despite different staining affinities (basophilic vs. eosinophilic). Erythroblast errors concentrate in ig (0.6%) and neutrophil (4.5%), consistent with nuclear maturation similarities between late-stage erythroblasts and early granulocytes. Monocyte's most frequent confusion involves ig (14.8%).

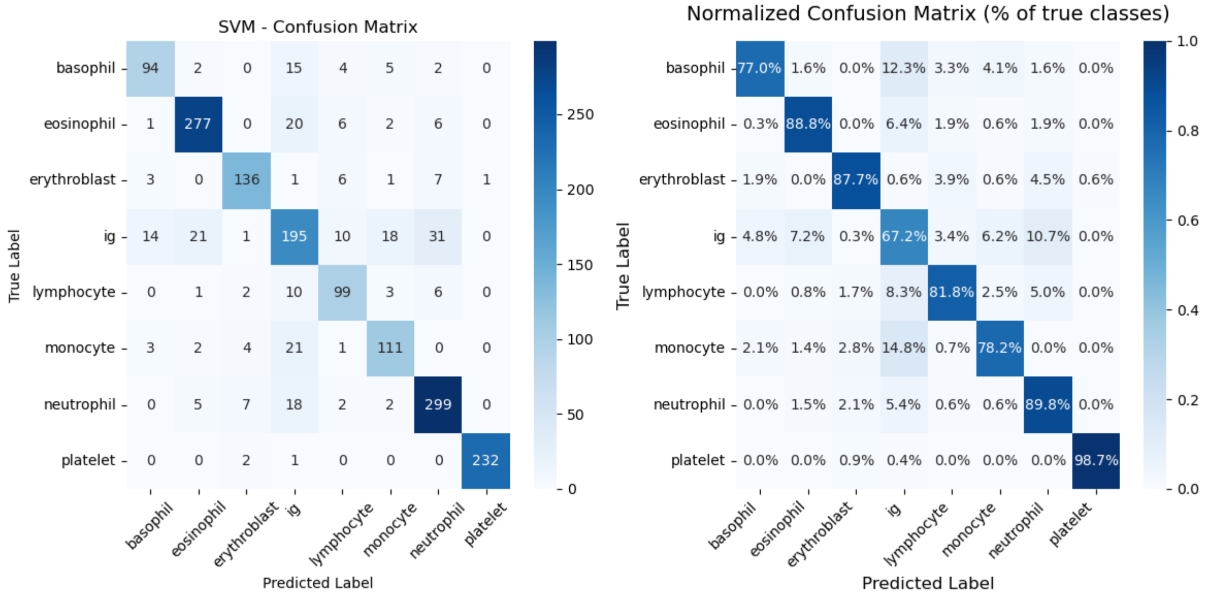

Figure 11: Confusion matrix of best SVM model showing absolute classifications (left) and normalized classifications per class (right).

Platelet exhibits zero confusion with lymphocyte, monocyte, or neutrophil (0.0% off-diagonal entries in platelet row), confirming complete separation in feature space. The asymmetric confusion pattern – ig frequently misclassified as neutrophil (10.7%) but neutrophil rarely as ig (5.4%) – suggests that mature granulocyte features are more discriminative than transitional ig characteristics, aligning with Random Forest's emphasis on chromatic variance (R/G_std) over shape descriptors that would better differentiate developmental stages.

The diagonal dominance (77.0–98.7% correct classification per class) validates the 15-feature set's adequacy for baseline performance, though confusion between certain cells types (basophil/eosinophil/ig/neutrophil) indicates potential gains from additional texture features targeting granule morphology or nuclear segmentation patterns.

4.4.3 Per Class Performance Curves

Precision-Recall Analysis

Per-class precision-recall curves (Fig. 12) quantify the trade-off between positive predictive value and sensitivity across varying decision thresholds, revealing performance heterogeneity driven by class imbalance and morphological ambiguity. Platelet achieves perfect discrimination (AUC-PR: 1.00) with precision maintained at 1.0 across all recall levels, reflecting complete feature-space separation confirmed by confusion matrix and PCA clustering. Eosinophil (AUC-PR: 0.97), erythroblast (0.96), and neutrophil (0.93) demonstrate robust performance with curves maintaining >0.90 precision until ~0.80 recall, indicating that high-confidence predictions (top 80% probability scores) exhibit minimal false positives.

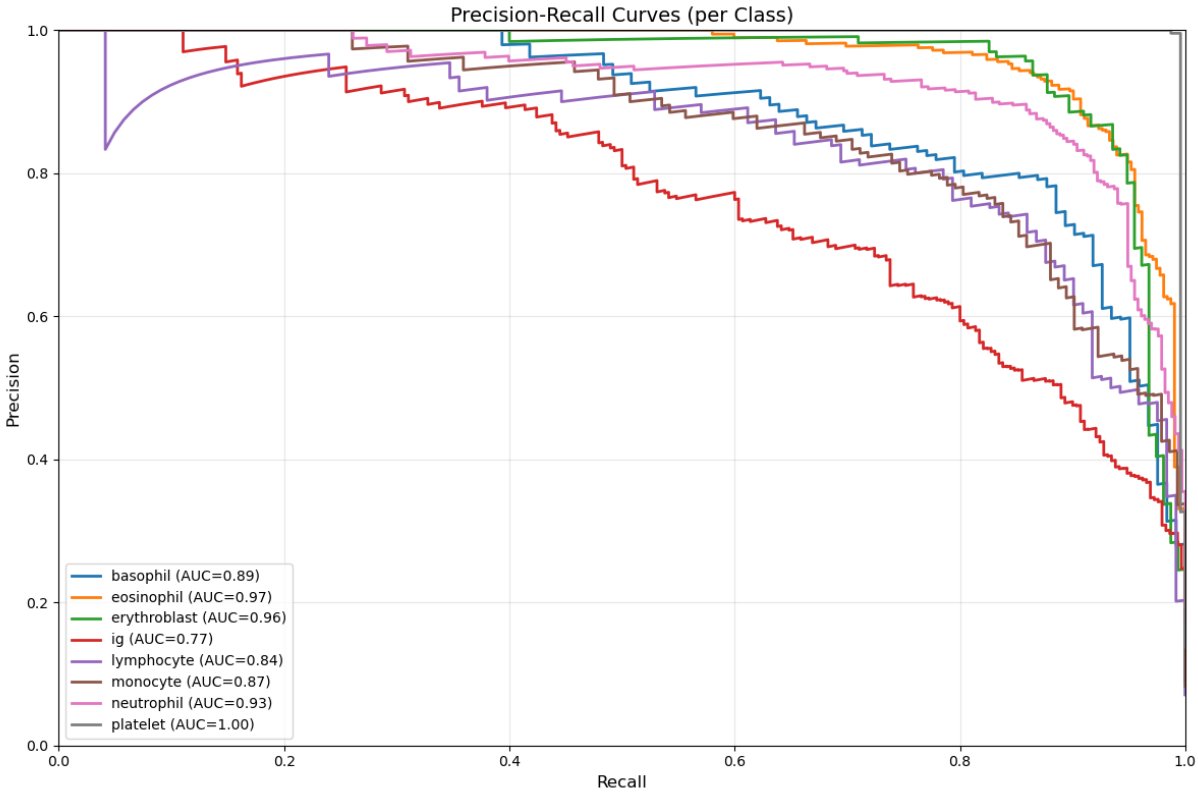

Figure 12: Precision-Recall curves per cell type class.

In contrast, immature granulocyte exhibits the lowest AUC-PR (0.77), with precision degrading from 0.85 to 0.60 as recall increases from 0.3 to 0.7. This steep precision-recall trade-off reflects feature overlap with neutrophils and eosinophils: the classifier must lower confidence thresholds (increasing false positives) to capture additional true ig cells, consistent with the 32.8% error rate observed in confusion matrix. Basophil (AUC-PR: 0.89), lymphocyte (0.84), and monocyte (0.87) occupy intermediate positions, their moderate AUC-PR values correlating with partial cluster overlap in PCA space and shared agranulocyte/granulocyte characteristics.

ROC Curve Validation

Receiver Operating Characteristic curves (Fig. 13) confirm precision-recall findings through complementary true-positive-rate/false-positive-rate metrics. All classes achieve AUC-ROC ≥0.93, with platelet (1.00), basophil (0.99), eosinophil (0.99), and erythroblast (0.99) reaching near-perfect discrimination. The uniformly high AUC-ROC values (0.93–1.00, range: 0.07) appear compressed relative to AUC-PR spread (0.77–1.00, range: 0.23), reflecting ROC's sensitivity to true-negative-rate dominance in multi-class one-vs-rest formulations where negative samples outnumber positives.

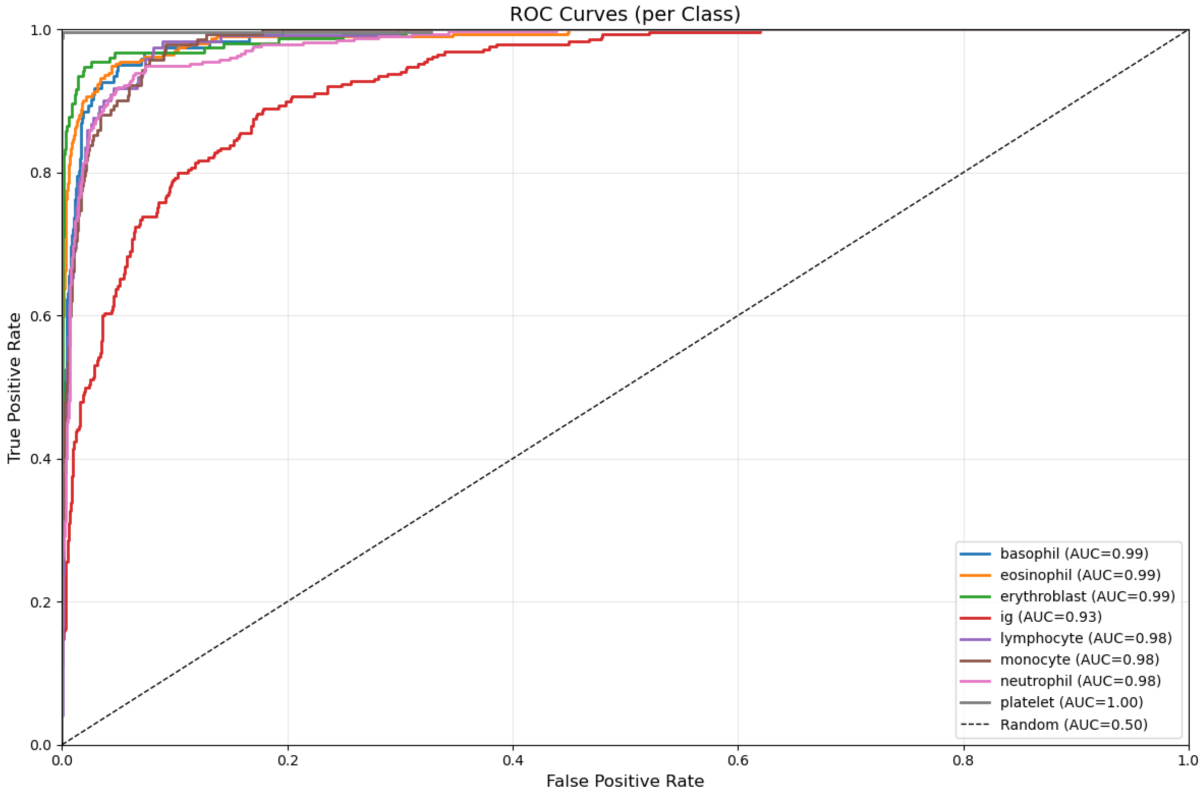

Figure 13: ROC curves per cell type class.

Immature granulocyte's ROC performance (AUC: 0.93) substantially exceeds its precision-recall performance (AUC: 0.77), demonstrating the metric's optimism under class imbalance: high true-negative rates from correctly excluding 87.6% of non-ig samples inflate AUC-ROC, whereas precision-recall directly penalizes the 95 false positives among 290 ig predictions. The 0.16-point AUC discrepancy (ROC: 0.93 vs. PR: 0.77) quantifies imbalance-induced metric divergence, validating precision-recall curves as the primary performance indicator for this moderately imbalanced dataset.

All ROC curves exhibit steep initial ascent (>0.90 TPR at <0.10 FPR), confirming that the SVM's decision boundaries effectively separate classes despite feature overlap, with misclassifications concentrated in ambiguous boundary regions rather than distributed randomly across feature space.

4.4.4 Sample Size vs. F1-Score Analysis

The relationship between training sample size and F1-score (Fig. 14) reveals that classification performance is primarily driven by class-specific feature discriminability rather than sample abundance. The near-flat trendline demonstrates negligible correlation between training support and F1-score, indicating that the moderately imbalanced dataset (7.1–19.5% per class) does not systematically disadvantage minority classes.

Figure 14: Relationship between training sample size and F1-score.

Platelet achieves perfect performance (F1: 0.99) despite moderate training support (n=1,883, 13.8%), validating that distinctive morphological features – small size, high circularity, uniform cytoplasm – enable robust classification independent of sample quantity. Similarly, eosinophil (n=2,496, 18.3%, F1: 0.89) and neutrophil (n=2,664, 19.5%, F1: 0.87) demonstrate that high F1-scores correlate with feature uniqueness rather than majority-class advantage, as evidenced by erythroblast's strong performance (F1: 0.89) at only n=1,240 (9.1%).

Conversely, immature granulocyte exhibits the lowest F1-score (0.68) despite substantial training support (n=2,320, 17.0%), ranking third in sample size yet seventh in performance. This inverse relationship confirms that ig's poor classification stems from inherent morphological ambiguity – transitional developmental features overlapping with mature granulocytes – rather than insufficient training data. Similarly, lymphocyte (n=969, 7.1%, F1: 0.80) and monocyte (n=1,136, 8.3%, F1: 0.78) perform comparably to basophil (n=978, 7.2%, F1: 0.79) despite similar minority-class status, further demonstrating that feature separability, not sample abundance, determines classification success.

The absence of sample-size dependency suggests that data augmentation or oversampling strategies would yield minimal performance gains; instead, improvements require enhanced feature engineering targeting granule morphology, nuclear segmentation patterns, or chromatin texture to better differentiate morphologically similar classes.

4.4.5 Class-Pair Confusion Analysis

The symmetric confusion matrix (Fig. 15) reveals that immature granulocyte accounts for the majority of class-pair errors: ig-neutrophil (49), ig-eosinophil (41), and ig-monocyte (39) collectively represent 129/268 total confusions (48.1%), reflecting transitional developmental features overlapping with multiple myeloid lineages. Eosinophil-neutrophil (11) and basophil-ig (29) confusions stem from shared granulocyte characteristics – lobulated nuclei, cytoplasmic granulation – where staining intensity differences prove insufficient for complete separation. Platelet exhibits near-zero confusion (4 total errors), validating complete feature-space isolation.

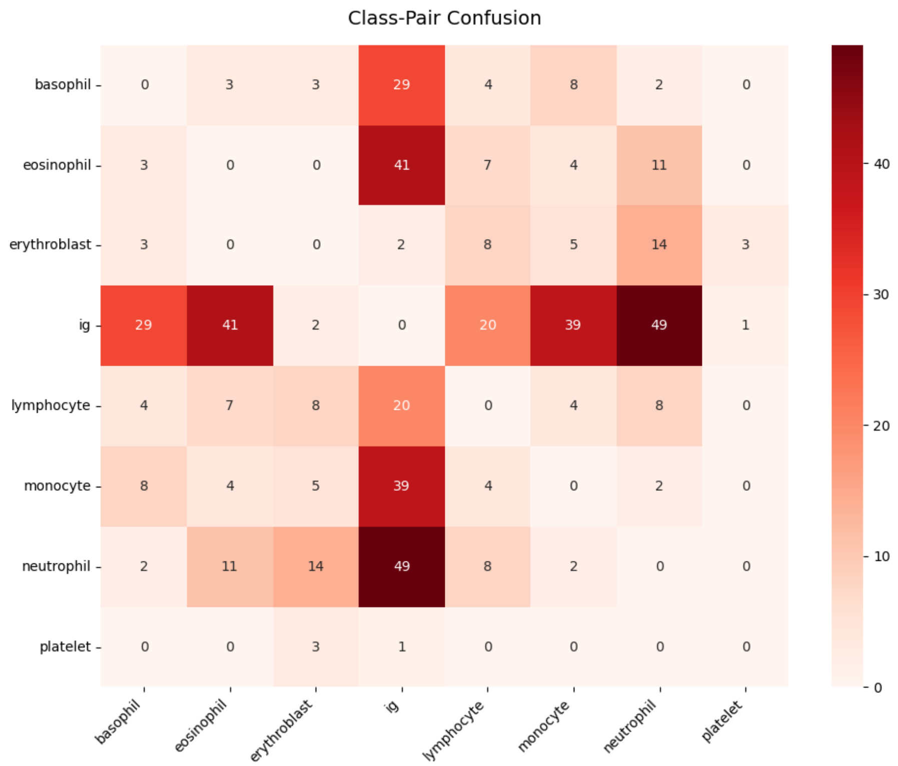

Figure 15: Class-Pair confusion matrix.

4.5 Conclusion and Outlook

The SVM baseline achieves 84.4% test accuracy with 15 handcrafted features, demonstrating robust performance yet revealing limitations in distinguishing morphologically similar cells. Systematic confusion among basophil/eosinophil/ig/neutrophil reflects handcrafted features' inability to capture subtle developmental variations, as evidenced by texture features' limited discriminative contribution (16.8% RF importance) despite biological relevance. Convolutional

Neural Networks offer potential for 5–15 percentage point accuracy improvements through hierarchical feature learning that automatically discovers multi-scale morphological patterns – nuclear chromatin texture, granule distribution, cytoplasmic heterogeneity – beyond manually engineered descriptors, potentially approaching higher performance (>95%) for automated hematological diagnostics.